Four characterisations of an identity system over a pointed type are logically equivalent

The Univalent Foundations Program

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Editorial note. The counted statement is the author's thm environment at source lines 1152-1160 and its proof environment opens at line 1167; between them lies the author's prose remark about the relation to homotopy induction, which is not part of the counted unit and is not retained, so the retained proof range 1167-1193 is the whole proof environment.

Editorial note. The four item labels item:identity-systems1 to item:identity-systems4 are defined inside the retained statement and are exactly the conditions the retained proof refers to; they need no external entry.

Editorial note. The retained counter values (chapter 5, section 8, theorem offset 1) make the counted theorem print as Theorem 5.8.2, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:identity-systems = 5.8.2).

Editorial note. Two labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: thm:contr-paths (book Theorem 3.11.8, defined at line 1057 of the logic chapter's pinned file) and thm:total-fiber-equiv (book Theorem 4.7.7, defined at line 767 of the equivalences chapter's pinned file). Both are cited for one step each inside an otherwise complete argument.

Editorial note. The retained preamble is the book's own macro library and layout adapter as stored for this chapter (retained-preamble.tex, the file's full length read at generation time); no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the logical equivalence of the four characterisations of an identity system (author label thm:identity-systems, book Theorem 5.8.2).

Allowed standard prerequisites (editorial).

- Pointed predicate and pointed map. A pointed predicate over (A,a_0) is a family R : A \to \type together with r_0 : R(a_0), and a pointed map is a family of maps respecting the base points.. They are the objects among which the four conditions are stated (source lines 1152-1155).
- Identity system at a point. R with r_0 is an identity system at a_0 when for every D : \prd{b:A} R(b) \to \type and d : D(a_0,r_0) there is f : \prd{b:A}{r:R(b)} D(b,r) with f(a_0,r_0) = d.. It is condition 1 of the counted theorem and the hypothesis of the first implication.

- Contractibility of a type of pointed maps. Contractibility is the combination of inhabitation and the equality of any two of its elements, obtained here through function extensionality and the path types of \Sigma-types.. Condition 2 and the derivation of condition 3 from it (source lines 1175-1185).
- Transport and \Sigma-types preserving equivalences. Transport in a family of types is an equivalence, and a \Sigma-type formed over an equivalence is again an equivalence.. Condition 3 is turned into condition 4 with these two facts (source line 1188).

Source reference. Original theorem 3.11.8: sources/hott-logic/pinned-original-source.tex, source lines 1057–1057. Book Theorem 3.11.8, that the path type of a contractible type is contractible, is cited only at the step where condition 3 is turned into condition 4. Navigation pointer only: the counted theorem's own proof is retained in full, and the pointer span is the line of the label in the pinned file that carries its statement (the fact is recalled in the retained text).

Source reference. Original theorem 4.7.7: sources/hott-equivalences/pinned-original-source.tex, source lines 767–767. Book Theorem 4.7.7, in its 'if' direction, that \Sigma-types respect equivalences, is cited only for one step of the third implication. Navigation pointer only: nothing of the counted argument is replaced by it, and the pointer span is the line of the label in the pinned file that carries its statement.

Theorem 5.8.2. *For a pointed predicate (R, r_0) over (A, a_0), the following are logically equivalent.*

(i) (R, r_0) *is an identity system at* a_0.
(ii) For any pointed predicate (S, s_0)*, the type* $\mathsf{ppmap}(R, S)$ *is contractible.*
(iii) For any $b : A$*, the function* $\mathsf{transport}^R(-, r_0) : (a_0 =_A b) \to R(b)$ *is an equivalence.*
(iv) The type $\sum_{(b:A)} R(b)$ *is contractible.*

Proof. First, assume (i) and let (S, s_0) be a pointed predicate. Define $D(b, r) :\equiv S(b)$ and $d :\equiv s_0 : S(a_0) \equiv D(a_0, r_0)$. Since R is an identity system, we have $f : \prod_{(b:A)} R(b) \to S(b)$ with $f(a_0, r_0) = s_0$; hence $\mathsf{ppmap}(R, S)$ is inhabited. Now suppose $(f, f_r), (g, g_r) : \mathsf{ppmap}(R, S)$, and define $D(b, r) :\equiv (f(b, r) = g(b, r))$, and let $d :\equiv f_r \cdot {g_r}^{-1} : f(a_0, r_0) = s_0 = g(a_0, r_0)$. Then again since R is an identity system, we have $h : \prod_{(b:A)} \prod_{(r:R(b))} D(b, r)$ such that $h(a_0, r_0) = f_r \cdot {g_r}^{-1}$. By function extensionality and the characterization of paths in Σ-types and path types, these data yield an equality $(f, f_r) = (g, g_r)$. Hence $\mathsf{ppmap}(R, S)$ is an inhabited mere proposition, and thus contractible; so (ii) holds.

Now suppose (ii), and define $S(b) :\equiv (a_0 = b)$ with $s_0 :\equiv \mathsf{refl}_{a_0} : S(a_0)$. Then (S, s_0) is a pointed predicate, and $\lambda b.\, \lambda p.\, \mathsf{transport}^R(p, r) : \prod_{(b:A)} S(b) \to R(b)$ is a pointed family of maps from S to R. By assumption, $\mathsf{ppmap}(R, S)$ is contractible, hence inhabited, so there also exists a pointed family of maps from R to S. And the composites in either direction are pointed families of maps from R to R and from S to S, respectively, hence equal to identities since $\mathsf{ppmap}(R, R)$ and $\mathsf{ppmap}(S, S)$ are contractible. Thus (iii) holds.

Now supposing (iii), condition (iv) follows from original theorem 3.11.8, using the fact that Σ-types respect equivalences (the "if" direction of original theorem 4.7.7).

Finally, assume (iv), and let $D : \prod_{(b:A)} R(b) \to \mathcal{U}$ and $d : D(a_0, r_0)$. We can equivalently express D as a family $D' : (\sum_{(b:A)} R(b)) \to \mathcal{U}$. Now since $\sum_{(b:A)} R(b)$ is contractible, we have

$$p : \prod_{u:\sum_{(b:A)} R(b)} (a_0, r_0) = u.$$

Moreover, since the path types of a contractible type are again contractible, we have $p((a_0, r_0)) = \mathsf{refl}_{(a_0,r_0)}$. Define $f(u) :\equiv \mathsf{transport}^{D'}(p(u), d)$, yielding $f : \prod_{(u:\sum_{(b:A)} R(b))} D'(u)$, or equivalently $f : \prod_{(b:A)} \prod_{(r:R(b))} D(b, r)$. Finally, we have

$$f(a_0, r_0) \equiv \mathsf{transport}^{D'}(p((a_0, r_0)), d) = \mathsf{transport}^{D'}(\mathsf{refl}_{(a_0,r_0)}, d) = d.$$

Thus, (i) holds. □